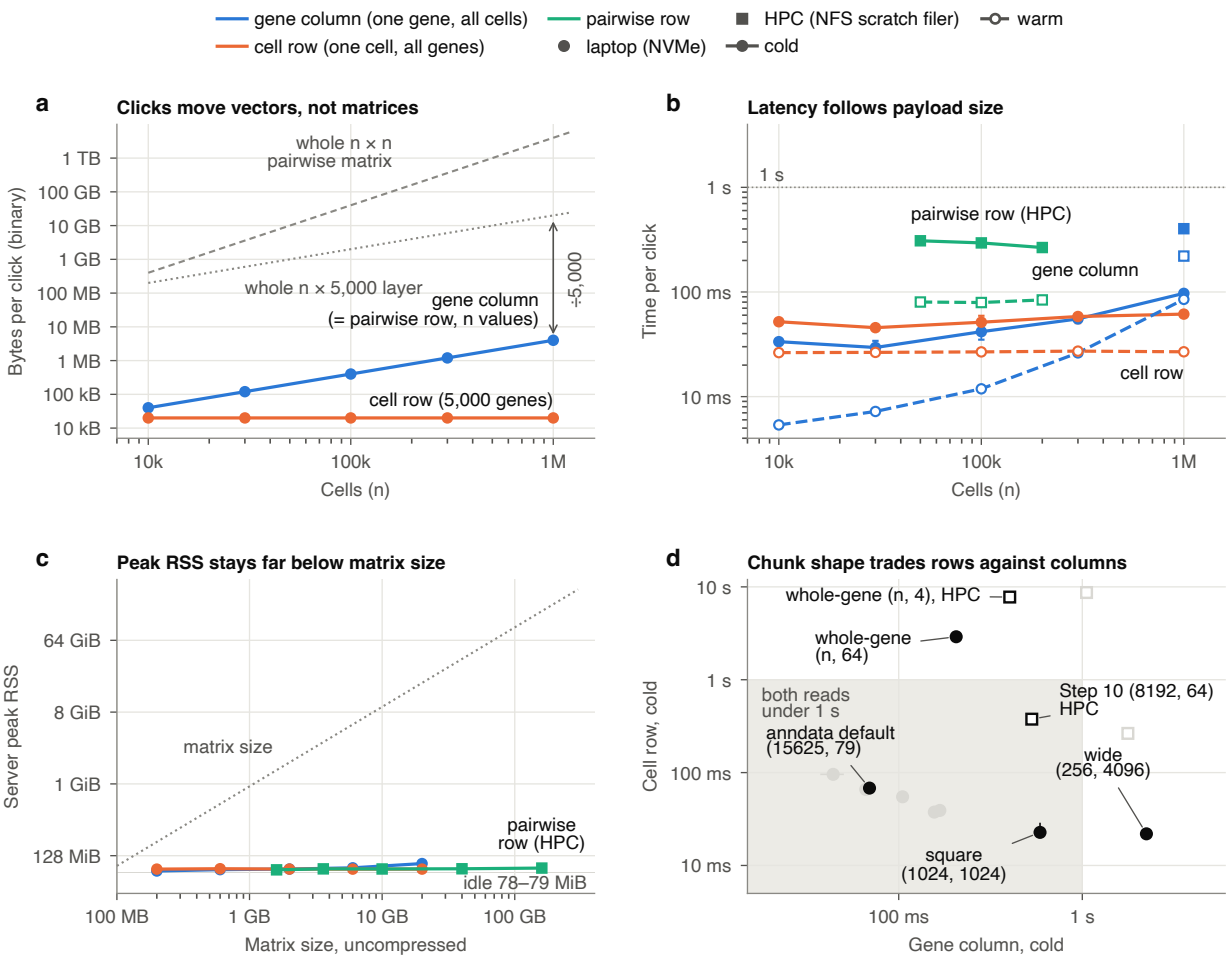
gene column (one gene, all cells)
cell row (one cell, all genes)
pairwise row
laptop (NVMe)
HPC (NFS scratch filer)
cold
warm
a
Clicks move vectors, not matrices
whole n × n
pairwise matrix
whole n × 5,000 layer
gene column
(= pairwise row, n values)
-5,000
cell row (5,000 genes)
Bytes per click (binary)
1 TB
100 GB
10 GB
1 GB
100 MB
10 MB
1 MB
100 kB
10 kB
10k
100k
1M
Cells (n)
b
Latency follows payload size
1 s
pairwise row (HPC)
gene column
cell row
Time per click
1 s
100 ms
10 ms
10k
100k
1M
Cells (n)
c
Peak RSS stays far below matrix size
matrix size
pairwise
row (HPC)
idle 78–79 MiB
Server peak RSS
64 GiB
8 GiB
1 GiB
128 MiB
100 MB
1 GB
10 GB
100 GB
Matrix size, uncompressed
d
Chunk shape trades rows against columns
whole-gene (n, 4), HPC
whole-gene
(n, 64)
both reads
under 1 s
anndata default
(15625, 79)
Step 10 (8192, 64)
HPC
wide
(256, 4096)
square
(1024, 1024)
Cell row, cold
10 s
1 s
100 ms
10 ms
100 ms
1 s
Gene column, cold